

## Basic course

KNX system arguments

KNX Association

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## **Preface**

This training document is created for the Basic course. Sections with a heading marked [E] contain important information that is also required for the NextGen course. This includes any sub-sections within them. If a heading does not have [E], it means the content is only for reference and is not mandatory for the NextGen course.

The documentation focuses on the most used configuration mode in KNX, i.e., S-Mode (System Mode). Configuration modes that used to be supported in the past like E-Mode (Easy Mode) are not covered further.

## 1 KNX Association: A brief outline [E]



Figure 1: KNX – The history

The KNX Association was established in 1990 in Brussels, Belgium, originally founded as the "EIB Association" (EIBA) by 15 members.

Its primary objective was to promote smart homes and buildings in general, with a particular focus on the EIB system, which was developed jointly by several renowned manufacturers. In 1999, the EIBA merged with two other European associations:

- BCI (France), which promoted the Batibus system.
- The European Home Systems Association (The Netherlands), which promoted the EHS system.

By 2006, it was decided to promote a single system, KNX. Following this merger, the association was renamed "KNX Association."

The KNX Association is responsible for:

- Promoting the KNX technology.
- Defining a new open standard, "KNX™," for smart homes and buildings.
- Establishing the KNX Trademark as a symbol of quality and multi-vendor interworking.
- Positioning KNX as both a European and global standard.

Since KNX is backward compatible with EIB, most devices can be labelled with both the KNX and EIB logo.

## 2 Activities of KNX Association – International standardization [E]

The KNX Association is involved in the following activities:

- Continuing the technical development and promotion of the KNX standard in collaboration with member companies.
- Ongoing development of the common design and commissioning tool, ETS™.
- Sales and support of ETS via the MyKNX portal (<https://my.knx.org>).
- Granting the KNX trademark for compatible devices through product certification.
- Promoting KNX training by certifying training centers and providing globally standardized course materials.
- Engaging in national and international standardization activities.
- Supporting the formation of national groups of manufacturers and/or users worldwide.
- Promoting "scientific partnerships" with universities and technical institutes to introduce the KNX system to students and support research.
- Promoting the "NextGen" educational program in secondary schools, colleges and universities to introduce the KNX system to young apprentices.
- Offering technical support for manufacturers developing KNX solutions.
- Defining testing and quality standards in cooperation with KNX member companies, with official conformity tests conducted by KNX accredited test labs.
- Marketing activities, including the website, trade fairs, and brochures.

The KNX Association has now grown to over 500 members. A current membership list can be found at [www.knx.org](http://www.knx.org).



Figure 2: KNX in standardization.

At the end of 2003, the KNX Standard was approved by CENELEC (European Committee for Electrotechnical Standardization) as the European Standard for Home and Building Electronic Systems, under the EN 50090 Series. It was also approved by CEN, with EN 13321-1 covering media and protocol, and EN 13321-2 for KNXnet/IP.

By the end of 2006, KNX was recognized as a global standard under ISO/IEC 14543-3 (parts 1 to 7). In May 2013, KNX was approved as a Chinese standard (GB/T 20965), and it is also recognized in the USA under ANSI/ASHRAE 135.

In November 2019, KNX IP Secure was recognized as an international security standard (EN ISO 22510), covering open data communication for building automation and management via (secure) KNXnet/IP.

### 3 KNX vs conventional technology [E]

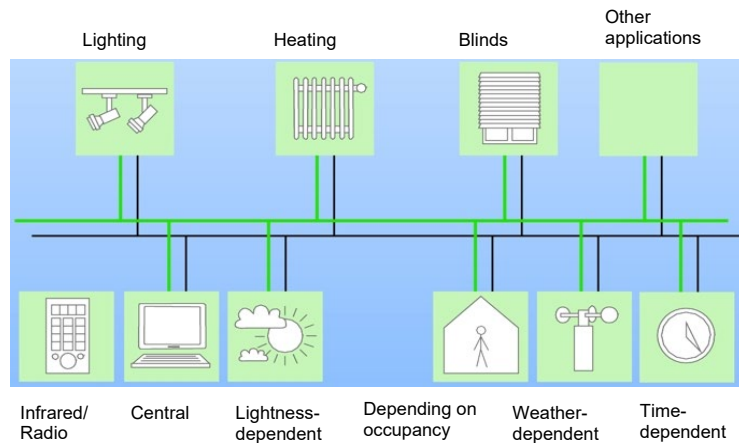


Figure 3: KNX – introduction to the technology

In the case of the most widely used medium, “Twisted Pair (TP),” a control cable can be installed parallel to the 230 V cable. This offers several advantages compared to conventional installations:

- **Reduced cabling:** The amount of cabling is significantly minimized when bus devices are arranged in a decentralized way.
- **Increased functionality:** The system can support a wider range of functions.
- **Improved transparency:** The installation becomes more organized and easier to understand.

This control cable:

- **Connects actuators,** e.g., loads and sensors e.g., push buttons.
- **Provides power:** It supplies power to the bus devices for their communication, and often also fully powers the devices (sensors, actuators, controllers).
- **Allows quick and simple wiring:** Just connect 2 cable cores, free branching, no terminating resistors, up to 1,000 m without repeaters.
- Ensures stable, uninterrupted communication.

Since all KNX bus devices are equipped with their own intelligence (microcontroller), there is no need for a central control unit (e.g., a PC). As a result, KNX is suitable for both small home installations and large commercial projects (such as hotels, airports, and office buildings).

## 4 KNX System specification

### 4.1 KNX Media [E]

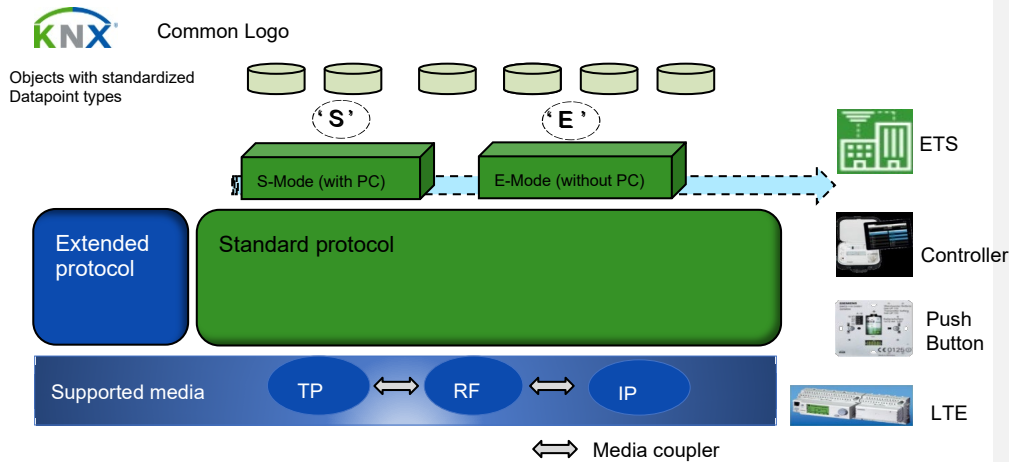


Figure 4: KNX System overview

As explained earlier, the exchange of KNX data between devices typically occurs through a separate control cable. However, KNX data can also be transmitted wirelessly via **KNX Radio Frequency (RF)** and through **Ethernet/Wi-Fi (KNX IP)**. Additionally, KNX telegrams can be transmitted over other media, such as optical fiber, using appropriate interfaces.

When different media are connected, media couplers must be used. The type of medium a device uses is indicated on its product label.



## 4.2 Areas of application for the various media [E]

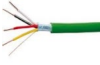
|                                                                                   | Medium    | Transmission via                        | Preferred areas of application                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | TP        | Separate control cable                  | <ul style="list-style-type: none"> <li>• New installations</li> <li>• Extensive renovations</li> <li>• Highest level of transmission reliability</li> </ul>                                                           |
|  | RF        | Radio line                              | <ul style="list-style-type: none"> <li>• When no cables can be installed</li> </ul>                                                                                                                                   |
|  | IP (IPv4) | Ethernet/Wi-Fi                          | <ul style="list-style-type: none"> <li>• In large installations where a fast backbone is needed</li> <li>• For communication with mobile devices</li> </ul>                                                           |
| <b>KNX IoT</b>                                                                    | IP (IPv6) | Thread/(single pair) Ethernet/Wi-Fi/... | <ul style="list-style-type: none"> <li>• All the above, depending on used medium (wired/wireless)</li> <li>• It can be used for new KNX installations as well as extensions of existing KNX installations.</li> </ul> |

Figure 5: Areas of application for the various media

## 4.3 Types of configuration [E]

Depending on what is marked on the product label, devices can be configured (i.e., linked logically and have their parameters set) through the following methods:

- **E-Mode (Easy Mode):**

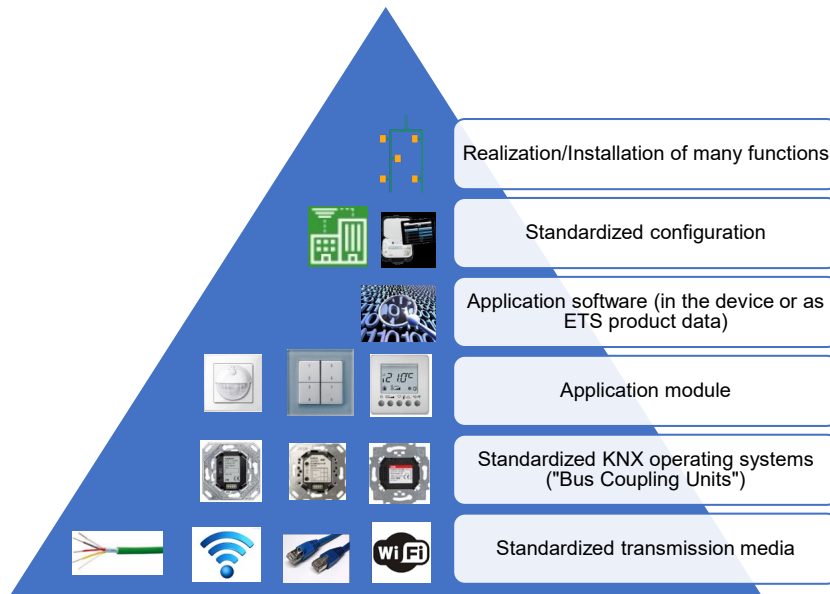
- Configuration is performed without a PC, using a central controller device, push buttons.
- This type of configuration is intended for skilled electrical installers with basic bus knowledge.
- E-Mode compatible devices typically have limited functionality and are designed for small to medium-sized installations.
- In E-Mode devices, the application program is pre-installed in the device.

- **System Mode (S-Mode):**

- The installation design and configuration are handled using the manufacturer-independent ETS software on a PC, with product data imported into ETS.
- This configuration method is intended for KNX-certified building designers and electrical installers and can be used for installations of any size.

Some devices support both **E-Mode** and **S-Mode**. For example, devices with the **LTE label** (mainly in HVAC applications) are usually configured using LTE (Logical Tag Extended) mechanisms. However, these devices also include a defined S-Mode interface, allowing them to be linked with S-Mode compatible devices.

## 4.4 KNX interworking [E]



**Figure 6: KNX interworking**

Devices from different KNX manufacturers and functional areas that bear the KNX trademark (i.e., the device is tested and certified) and use the same configuration mechanism can be integrated into a functioning installation, thanks to KNX standardization. This standardization applies to:

- **Telegrams:** Devices typically use standard telegrams for data transmission. In some cases, telegrams with extended length may be used to transmit larger amounts of data.
- **Useful data in telegrams:** For various functions such as switching, dimming, shutter control, and HVAC, KNX-certified devices must use predetermined data formats to ensure compatibility.

This standardization ensures that devices from different manufacturers can work together seamlessly in a KNX system.

## 4.5 KNX Secure

By following the guidelines provided in the KNX Security Checklist (available in the [Marketing Content Library](#) on support.knx.org), KNX installations can be largely protected against unauthorized manipulation. However, for certain media, such as RF and IP, and for some building types, additional security measures may be necessary.

To address this, **KNX Secure** was developed as an extension of the KNX system. It enables:

- **Secure access via IP (KNX IP Secure)**
- **Secure communication between individual devices (KNX Data Secure)**

Devices that support KNX Secure can be identified by their product label.

## 5 Success figures<sup>1</sup> [E]

- Millions of installed devices worldwide
- Thousands of KNX registered and certified devices
- Over 500 KNX members (manufacturers)
- More than 500 certified training centers
- Over 15 international, KNX accredited test labs
- Hundreds of thousands of implemented projects globally



Figure 7: Certified KNX training centres worldwide

<sup>1</sup> For current figures, please consult the KNX website ([www.knx.org](http://www.knx.org))

## 6 The unique advantages of KNX [E]

### 6.1 Interworking

One of the greatest advantages of KNX is the “**Interworking**” between devices from different KNX manufacturers. Since all KNX devices “speak” and “understand” the same language, they can communicate with each other without issues. For example:

- A presence detector from manufacturer A can work with the alarm system from manufacturer B.
- A room thermostat from manufacturer A can control the valves from manufacturer B.
- An “All off” button from manufacturer A can switch off lights controlled by switch actuators from manufacturers B, C, D, and others.

If a device from manufacturer A becomes defective, discontinued, or is no longer preferred, it can easily be replaced by a newer version of this device from manufacturer A or by a device from manufacturer B with similar functionality. Using the ETS software, the new device can be seamlessly integrated into the existing KNX installation.

For customers as well as for installers KNX provides

- long-term stability
- long-term maintainability
- long-term expandability
- independence from single suppliers

### 6.2 One single configuration tool

Using multiple home and building automation systems (e.g., separate systems for lighting, HVAC, and sun shading) often requires the system integrator to learn different software packages or configuration methods to install and manage these devices. Software provided by different manufacturers, or even for different devices from the same manufacturer, can vary greatly in terms of user interface and operation. This increases the training effort needed for the system integrator.

However, with KNX, the integrator only needs to be trained in one software tool, ETS, to configure KNX devices across various application domains and from different manufacturers. In ETS, product data from any KNX manufacturer can be imported from a central repository. Because ETS is backward compatible, the integrator can import product data that were initially released for much older generations of ETS into the latest one. This means projects created years ago can still be imported, modified or extended using the most recent ETS version. A significant benefit of using one tool is the ability to fully document which functions of the installed devices are linked, especially for larger installations. This provides both the system integrator and the home and building owner with complete documentation of their installation.

The ETS software is available in four different versions, each tailored to specific target groups:

- **ETS Professional:** Designed for professional integrators and used on a desktop or laptop (Windows). It allows for the creation of an unlimited number of projects with no restrictions on the number of devices.
- **ETS Lite:** Limited to projects with up to 20 devices. This version is primarily intended for certified training centers.
- **ETS Bundle:** Provides three ETS Professional licenses at a reduced price, specifically for KNX manufacturers.
- **ETS Home:** Allows for the creation of one project with a maximum of 64 KNX devices. This version is designed for tech-savvy individuals or homeowners who may collaborate with a skilled KNX installer.

### 6.3 Wide range of KNX devices from many manufacturers and large service network

Home and building owners have access to a wide range of devices from various manufacturers, allowing them to select those that best meet their specific needs in terms of functionality (e.g., push buttons with or without scene or temperature control) and design preferences. If owners wish to modify or expand their KNX installation, they can rely on a vast network of qualified KNX system integrators and installers. Even if their previous system integrator is no longer active, the KNX website ([www.knx.org](http://www.knx.org)) provides a directory of alternative KNX system integrators in their area.

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### 6.4 Backward compatibility

Throughout its more than 30 years of existence, the KNX system has maintained full compatibility, meaning that even the earliest installations can still be expanded using the latest devices from manufacturers.

### 6.5 Energy efficiency

Due to global climate change and the growing scarcity of resources, building regulations in recent years have increasingly prioritized sustainability, making the use of building automation essential. Building automation offers several key benefits:

- **Integration of different systems:** Applications such as lighting, sun shading, and heating/cooling are fully synchronized, ensuring that they work together efficiently rather than independently or in conflict.
- **Adaptation to external climate conditions:** Systems can be optimized based on real-time climate data, using weather stations that measure external temperature, wind direction, wind speed, light intensity, and the position of the sun.
- **Cross-domain sensor interaction:** Sensors in one domain (e.g., a presence detector) can influence systems in other domains, such as lighting and temperature control.
- **Localized control:** Rooms or areas within a building can be individually controlled, offering greater efficiency compared to centralized control systems.
- **Energy consumption measurement:** Consumption can be measured at the functional level (e.g., using KNX energy actuators) rather than just at a global level for the entire building.
- **Energy management:** KNX supports the management of energy flows between local energy sources such as a Photovoltaic system (PV), local consumers (like heat pumps, electric vehicle charging stations, and household appliances), and the power grid.

These measures significantly reduce energy consumption, making investment in building automation highly cost-effective, with a strong return on investment.

## 6.6 Higher degree of convenience

In cases where a home or building owner wants to automate and electrify many different functions, KNX is more cost-efficient and easier to install than other conventional systems. With simple adjustments to the KNX installation, a system integrator can easily meet the changing needs of the owner. For example:

- When a new tenant in an office building has different electrical requirements than the previous tenant.
- When a private home needs to be converted into a nursing home as its residents age or become less mobile.
- When the home or building owner wants to enhance security by adding more sensors (e.g., glass-break or window sensors) or presence simulation devices.

KNX's flexibility allows for easy adaptation to these evolving demands, making it a future-proof solution.

## 7 KNX: Application examples [E]

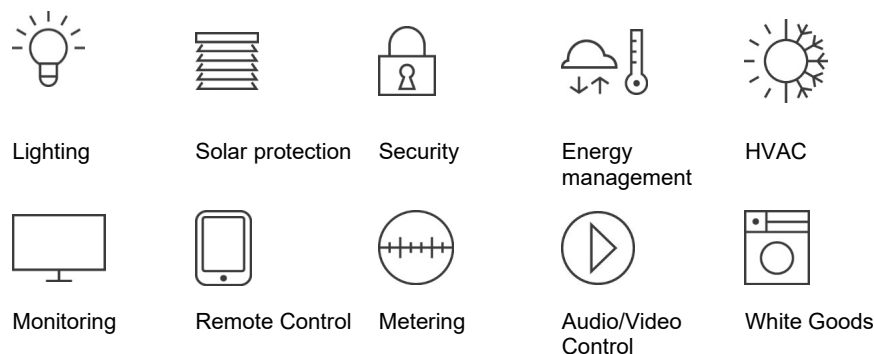


Figure 8: Possible application fields

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|   | <p><b>Example 1:</b> Implementation of central functions. When leaving the building, a single command can be used to control multiple functions simultaneously through KNX. This includes switching off all lights, the water supply, and specific sockets (such as for an electric oven) and reducing the level of heating or cooling. Additionally, the KNX alarm system can be activated, and the blinds can be adjusted based on the time of day. This centralized control enhances convenience, security, and energy efficiency.</p>                  |
|  | <p><b>Example 2:</b> In conference rooms, theaters, and living rooms, different lighting scenes can be activated depending on the activity, and users can modify these scenes at any time. For example, in administration buildings, energy savings of up to 75% can be achieved by implementing constant light control, using just one brightness sensor per building facade to adjust lighting based on natural light levels.</p>                                                                                                                        |
|  | <p><b>Example 3:</b> The status of all equipment in a flat can be displayed in clear text and controlled using fixed display units, or mobile devices such as smartphones or tablets. Similarly, in larger installations, this can be achieved using PCs and visualization software (a KNXnet/IP interface or KNXnet/IP router is required in this case). These visualization programs are available from various KNX manufacturers as well as independent software providers, offering flexible solutions for monitoring and controlling KNX systems.</p> |

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>Example 4:</b> By interfacing a KNX installation with the public communication network, the user can influence or query the building management functions (e.g., the heating) using a mobile phone. Alarm signals can be automatically routed to any required telephone number. KNX installations can also be remotely serviced and configured by the installer using any available media (e.g., the Internet). The time required for maintaining the building management system is thus considerably reduced.</p>                                                                   |
|    | <p><b>Example 5:</b> A large conference room can be divided into several independent areas when needed. By inserting partition walls, the KNX installation automatically detects and adjusts the assignment of switches and lights for each room section. This eliminates the need to modify the existing cabling, offering flexibility without additional wiring changes.</p>                                                                                                                                                                                                             |
|    | <p><b>Example 6:</b> Any number of panic switches (such as those that activate all lights) can be installed in a KNX system. Additionally, at night, a button can be pressed to activate the lights between the children's bedroom and the bathroom, with the lights automatically turning off after a set period. This provides both security and convenience for the user.</p>                                                                                                                                                                                                           |
|  | <p><b>Example 7:</b> KNX allows for individual room control of heating and cooling systems, with customizable heating and cooling profiles for each room. When a window is opened, the system automatically adjusts the heat or cold input for that room, resulting in energy savings of over 30% per year compared to a manually controlled system. Additionally, heat generation is regulated based on the actual demand from individual rooms, ensuring that heat is only produced when needed (e.g., when combined with presence detectors), further optimizing energy efficiency.</p> |
|  | <p><b>Example 8:</b> KNX allows for presence simulation during the absence of the building owner, mimicking normal daily activities such as lighting and blinds operation. This enhances security by creating the appearance that the building is occupied, deterring potential intruders.</p>                                                                                                                                                                                                                                                                                             |
|  | <p><b>Example 9:</b> The power consumption of individual circuits can be monitored using energy sensors or energy actuators. A KNX load management system can then temporarily switch off loads when predefined thresholds are exceeded. When integrated with a gateway to Smart Metering devices or renewable energy sources, the system ensures optimal use of self-generated energy. This is particularly useful when combined with photovoltaic generation and/or battery-powered electric vehicles, allowing efficient energy management and sustainability.</p>                      |





**Example 10:** Smart virtual assistants that support voice control, such as Amazon Alexa, Google Home or Siri, can be integrated with a KNX installation. This allows homeowners to control various functions—such as lowering the shutters, switching lights on or off, dimming them, or adjusting the temperature—simply through voice commands, offering enhanced convenience and automation.

## 8 Selling the benefits

When consulting with a potential customer, the electrical contractor or designer should focus on the **benefits** of KNX and address the customer's specific needs, rather than emphasizing technology or costs upfront. The goal is to provide a **system quotation**, not just a list of components. The process should proceed as follows:

- **Discuss the system quotation** with the customer, highlighting how it meets their needs and delivers value.
- **If the system price is too high**, consider redesigning certain functions (e.g., switching instead of dimming) to fit the budget.
- **In extreme cases**, reduce the installation cost by using more cost-effective components, such as:
  - Push button interfaces connected to conventional push buttons.
  - Push button BCUs.
  - Multi-functional combo actuators, which can handle switching, dimming, and/or sun shading control.

This approach helps align the system with the customer's expectations and budget, while still offering the core benefits of KNX.

## 9 KNX Partnership

Becoming a KNX partner is reserved for individuals who have successfully completed a full KNX Basic course (or a Condensed Practical Basic course) at a certified training center and passed both theoretical and practical exams.

Once certified as a KNX partner, the individual and their company will be featured on a global KNX website, which attracts thousands of international visitors each month. This increased visibility can lead to numerous new business opportunities and potential clients.

### 9.1 How to sign up for a course?

You can sign up for a certified KNX Basic course via the following link:

<https://www.knx.org/knx-en/for-professionals/get-started/get-certified/index.php>.

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### 9.2 Benefits of becoming a KNX Partner

A KNX Partner enjoys several advantages:

- **Official Partner Certificate:** This certificate serves as a clear indicator to potential customers of the partner's expertise and the quality of his/her work.
- **KNX Partner logo:** Partners receive the logo in electronic format, which can be used on promotional materials, websites, business cards, and other business-related documents, signaling a mark of quality.
- **ETS6 Lite license:** Upon acceptance into a Basic course by a certified KNX training center, students receive a time-limited cloud-based ETS6 Lite license.
- **ETS6 Professional upgrade voucher:** After passing the exams, partners receive a rebate voucher to upgrade their time-limited ETS6 Lite license to ETS6 Professional.
- **Increased visibility:** KNX partners can have their contact details featured on the international KNX website, offering exposure to thousands of potential clients and partners worldwide.

## 10 KNX Course offerings

KNX certified training centres offer several standardized courses in addition to the Basic course. These include the **Advanced course**, **HVAC Specialist course**, **Energy Management course**, **Tutor course**, and **Refresher course**. These courses enhance your skills and knowledge in specific KNX domains. You can enroll for a course at: [KNX Training Registration](#).

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|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KNX Advanced course          | <p>This course covers among others advanced topics such as:</p> <ul style="list-style-type: none"> <li>• Telegram structure</li> <li>• Flags</li> <li>• Interworking</li> <li>• KNX Couplers</li> <li>• Logic operations</li> <li>• IP Communication</li> <li>• ...</li> </ul> <p>The course concludes with a theoretical and practical exam.</p>                                                                                                                                                                                                                                                     |
| KNX HVAC Specialist course   | <p>This course is designed for KNX professionals working with devices impacting HVAC systems, which are often installed by different companies. This course ensures proper coordination between the KNX installer and the HVAC installer by providing knowledge on:</p> <ul style="list-style-type: none"> <li>• Heat distribution systems</li> <li>• Heating/cooling control</li> <li>• Domestic water heating</li> <li>• Background on humidity, dew point and mold prevention</li> <li>• Energy efficiency with KNX</li> <li>• ...</li> </ul> <p>The course concludes with a theoretical exam.</p> |
| KNX Energy Management course | <p>The Energy Management course highlights the growing importance of building automation in achieving energy efficiency, especially in response to evolving environmental and regulatory requirements. In this course, KNX partners are introduced to a variety of KNX-based strategies and tools designed for effective energy management. These include methods for visualizing energy consumption to raise user awareness and the integration of smart meters to enable advanced features such as load management.</p> <p>The course concludes with a theoretical exam.</p>                        |
| KNX Tutor course             | <p>This course is aimed at partners who wish to establish a certified KNX training centre, or work as a KNX tutor at an existing certified training centre.</p> <ul style="list-style-type: none"> <li>• This course covers a range of topics, such as:</li> <li>• Serial data transmission and KNX protocol</li> <li>• Microcontrollers</li> <li>• Application program</li> <li>• Structure and Scope of KNX Association</li> <li>• ...</li> </ul> <p>The course concludes with a theoretical exam.</p>                                                                                              |
| KNX Refresher course         | <p>This course is intended for KNX partners who completed the Basic course a long time ago. This course helps participants refresh and</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Basic course

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|  |                                                                                                                                               |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------|
|  | further develop their KNX skills by providing the latest information about the KNX system and tool updates. There is no exam for this course. |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------|

## 11 KNX Virtual [E]

KNX Virtual is a free Windows-based application that simulates a KNX installation, allowing students to explore and interact with KNX technology without the need for physical hardware. The commissioning is performed using ETS. It offers a practical environment to learn the basics of KNX and gain confidence before starting real-world projects.

More than 10 different types of KNX devices are represented, all connected to a single TP line. These virtual devices interact with various building loads, such as:

- Lamps
- Blinds
- Heating & cooling valves

KNX Virtual also allows users to experiment with more sophisticated building functionalities, including:

- Weather modules
- Alarms
- Scenes
- Logic operations

You can easily download KNX Virtual through your MyKNX account. Simply follow these steps:

1. Log in to your MyKNX account.
2. Go to the shop and add "KNX Virtual" to your shopping basket.
3. Complete the order process (you will not be charged).
4. After finishing the order, you will receive an email with a download link for the tool.